Personality and Mental Health

RESEARCH ARTICLE OPEN ACCESS

An Investigation of Longitudinal Measurement Invariance and Item Response Theory Analysis of the LPFS-BF 2.0 in a Large Clinical Norwegian Sample

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Received: 29 January 2025 | **Revised:** 4 June 2025 | **Accepted:** 19 June 2025

Keywords: alternative model for personality disorders | level of personality functioning | longitudinal measurement invariance | personality problems | self-report

ABSTRACT

The LPFS-BF 2.0 (Level of Personality Functioning—Brief Form 2.0) is a brief self-report instrument that was developed in order to gain a quick first impression of personality dysfunction and was designed to align with the Level of Personality Scale as outlined in the DSM-5 Alternative Model for Personality Disorders (AMPD). In the current study, we used a factor analytic framework to evaluate the longitudinal measurement invariance of the Norwegian version of the LPFS-BF 2.0 in a sample of 1030 patients referred for personality disorder treatment with the R package *lavaan*. Additionally, we estimated item response theory (IRT) parameters for the 12 items, as well as local reliability, in order to facilitate comparisons with other studies. We found support for partial measurement invariance. Three items showed different thresholds across time points. However, these differences were minor and did not have a substantial impact on the estimated factor scores. The discrimination parameters were lower than could be expected for an instrument used in a clinical setting. Taking these findings together, we conclude that although the LPFS-BF 2.0 seems to measure the same latent trait across time, the items do not discriminate well between patients with high and low scores in a Norwegian population of adults seeking treatment for PD. Important next steps would be to estimate IRT parameters for comparable patient samples from other countries to assess the generalizability of our findings and to compare the LPFS-BF 2.0 scores to scores obtained through a clinical interview that aligns with the AMPD.

1 | Introduction

For many years, personality disorders (PDs) were diagnosed using a categorical approach, which resulted in a patient being assigned to one or more PD categories. However, the field is moving toward a more continuous (dimensional) approach, which is reflected in the latest version of the ICD (ICD-11) as well as the Alternative Model for Personality Disorders in the DSM-5 (AMPD), both of which are increasingly being adopted in various countries. With this new approach, the focus has shifted from distinct PD categories to a global personality dysfunction score (which is conceptualized as the Level of Personality Functioning Scale in the AMPD), where specific traits can be scored in a second step in order to give a more nuanced interpretation of the patient's personality problems. Supporters of the continuous/dimensional approach underline

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the importance of having a good measure of personality dysfunction that can be used in treatment planning and evaluation of treatment outcome (Bach et al. 2015; Morey 2017; Morey et al. 2022).

These significant changes to the diagnostic paradigm prompt the need for new screening and assessment instruments. A highly popular self-report inventory, developed to screen for global personality dysfunction, is the Level of Personality Functioning Scale-Brief Form 2.0 (LPFS-BF 2.0). This instrument was originally developed in the Netherlands and is currently available in 24 languages (De Viersprong 2024). Its brevity (12 items each tapping into one of the 12 subdomains of the LPFS in the DSM-5) undoubtedly contributes to its popularity, as exemplified by the inclusion of the LPFS-BF 2.0 in the standard set of patient-reported outcomes for personality disorders (Prevolnik Rupel et al. 2021). Its construct validity and reliability have been assessed in both clinical and nonclinical groups and across various countries (Balzen et al. 2024; Łakuta et al. 2023; Le Corff et al. 2022; Natoli et al. 2022; Paap et al. 2024; Rossi and Diaz-Batanero 2024; Weekers et al. 2019, 2023; Wu et al. 2024). Taking the results of these studies together, it appears that although the LPFS-BF 2.0 shows a certain degree of multidimensionality, it can be considered as essentially unidimensional in several populations (e.g., Danish general population and Norwegian clinical population [Paap et al. 2024; Weekers et al. 2023]). However, the degree of multidimensionality present, as well as the quality of the items in terms of how well they discriminate among persons with different severity scores, do seem to differ somewhat across populations. For example, strong (0.66–0.90) correlations between the two factors were found in various studies (Bach and Hutsebaut 2018; Natoli et al. 2022; Paap et al. 2024), whereas a lower correlation of 0.44 was reported by Weekers et al. (2019).

Although originally the primary assessment goal of the LPFS-BF 2.0 was to gain 'a first impression of impairments in personality functioning', other assessment goals have also been explored, including measuring change in response to inpatient treatment (Weekers et al. 2019). Weekers et al. reported a high sensitivity to change, which would suggest that the instrument should be suitable for this goal. However, before measured change can be interpreted with confidence, it needs to be clarified whether the instrument shows longitudinal measurement invariance; that is, whether the factorial structure or dimensionality remains stable over time and whether the item parameters remain stable over time. Only if we find support for longitudinal measurement invariance can change scores be interpreted meaningfully. Although measurement invariance has been studied and at least partial support for the measurement invariance of a two-factor solution has been found across various language versions of the LPFS-BF 2.0 (Le Corff et al. 2022; Natoli et al. 2022), longitudinal measurement invariance has not yet been explored.

The main aim of this study is to evaluate longitudinal measurement invariance in a large clinical sample of Norwegian patients who completed the instrument at two time points (with a time interval of 6 months). We will take the unidimensional model (one factor) as a starting point, because this model shows the best alignment with the theoretical underpinnings of the LPFS as defined in the DSM-5. Additionally, we will estimate item response theory (IRT) models for each time point separately, to investigate reliability as a function of the latent trait (as measured with Fisher information) and strengthen the comparability of reported results with the first norming study for this instrument, which was conducted using the Danish version of the LPFS-BF 2.0.

2 | Method

2.1 | Participants

The current study is based on the same research database that was utilized in a recently published LPFS-BF 2.0 study conducted by our research group (Paap et al. 2024).[1] The database encompassed data gathered at 15 outpatient units within the Norwegian Network for Personality Disorders (The Network; Pedersen et al. 2022). All units are specialized in the assessment and treatment of personality disorders and personality-related difficulties. For a more detailed description of the patients, clinicians, treatment context and assessment procedures, we refer the reader to Kvarstein et al. (2023). The analyses in the current study are based on data collected at two time points, which we will refer to as T1 and T2. The data at T1 were gathered at the time of initial assessment, before admittance to treatment, from August 2017 to March 2024. The questionnaire was administered again after 6 months (T2). At T2, most patients had started therapy; 3% were still waiting for therapy to commence. Note that we only used data in our analyses that were collected from participants who had completed the LPFS-BF 2.0 at both T1 and T2 ($N = 1030$). See Table 1 for demographics and a more detailed description of diagnostic prevalences.

2.2 | Diagnostic Procedure

Diagnoses were made in accordance with the DSM-5 (American Psychiatric Association 2013), using the Mini International Neuropsychiatric Interview (MINI, Sheehan et al. 1994) for symptom disorders and the Structured Clinical Interview for DSM-5 Personality Disorders (SCID-5-PD) for section II PDs (First et al. 2016). Diagnostic reliability was not investigated. However, diagnostic assessments were performed in each unit by trained clinical staff and evaluated by a specialist in psychology/psychiatry. During the period the data used in this study were collected, training workshops were repeatedly conducted within Network units, focusing on personality disorders, diagnostic assessment, structured clinical interviews and principles of the LEAD procedure (Longitudinal, Expert, All-Data; Pedersen et al. 2013; Spitzer 1983). This means that the final diagnoses were based on all information that was available, which included referral letters, self-reported history and complaints and an overall clinical impression, in addition to information from diagnostic interviews. Further details on clinician training can be found in the study by Sveen et al. (2023), which is based on another subsample from the Network database.

TABLE 1 | Descriptive statistics for patients who completed the LPFS-BF 2.0 both at T1 and T2.

Variable, statistic	
Gender, *n* (%)	
Female	812 (79)
Male	218 (21)
Age, median (IQR)	28 (11)
Number of years of education after compulsory junior secondary school (ending at age of 16), median (IQR)	4 (3)
Living alone, *n* (%)	324 (32)
Symptom diagnoses,[a] *n* (%)	
One or more symptom disorders	949 (94)
Current major depressive disorder	341 (48)
Major depressive disorder in remission	58 (8)
Dysthymic disorder	62 (9)
Bipolar I	11 (2)
Bipolar II	36 (5)
Panic without agoraphobia	72 (11)
Panic with agoraphobia	67 (10)
Agoraphobia without panic	18 (3)
Social phobia	159 (23)
Obsessive–compulsive disorder	29 (4)
Generalized anxiety disorder	99 (15)
Post-traumatic stress disorder	81 (12)
Alcohol dependence/abuse	37 (5)
Substance dependence/abuse	18 (3)
Eating disorders	76 (11)
Personality disorders, *n* (%)	
One or more personality disorders	768 (88)
Paranoid PD	65 (7)
Borderline PD	311 (36)
Avoidant PD	381 (44)
Obsessive–compulsive PD	68 (8)
Dependent PD	48 (6)
PD NOS	112 (13)

Note: The percentages reported in this table are based on the total number of persons whose data were available for the given variable.
Abbreviation: Personality disorder not otherwise specified.
[a]Diagnoses occurring in less than 2% not included.

2.3 | The Level of Personality Functioning Scale Brief Form 2.0 (LPFS-BF 2.0)

The LPFS-BF 2.0 is scored using a four-point ordered response format. The scoring format has been shown to differ across studies, e.g., Dutch studies have been shown to use a scoring format from 1 to 4, whereas a scoring format from 0 to 3 was used in the Norwegian Network. In either case, the categories (ordered from low to high score) correspond to the following answering categories: 'Very False or Often False', 'Sometimes or Somewhat False', 'Sometimes or Somewhat True' and 'Very True or Often True'. In the current study, we chose to use the 1–4 scoring format to ensure easy comparability with other studies.[2]

The 12 items were developed such that they correspond to the 12 indicators of the LPFS. Items 1–6 were developed to measure self-functioning, and Items 7–12 cover interpersonal functioning. The LPFS-BF 2.0 was translated into Norwegian by a group of four clinical researchers, two of whom are fluent in both Dutch and Norwegian, following the guidelines outlined by Hambleton et al. (2004). More details regarding the translation procedure can be found in Paap et al. (2024).

In a Dutch sample of treatment-seeking adults, a Cronbach's alpha value of 0.82 was reported for the total score and somewhat lower values for the self-functioning (0.79) and interpersonal functioning (0.71) subscales (Weekers et al. 2019). A recent study conducted using the Danish version of the instrument reported a McDonald's ω value of 0.90 for the total score in a general population sample (Weekers et al. 2023). In a recent study conducted by our research team, we found a ω value of 0.71 for the self-dimension, 0.75 for the interpersonal dimension and 0.80 for the total scale in a large clinical Norwegian sample. Within a bifactor model framework, we also estimated omega-hierarchical values, which equaled 0.74 for the general factor, 0.42 for the self-specific factor and 0.02 for the Interpersonal specific factor (Paap et al. 2024). This indicates that a substantial amount of the reliable variance of the subscales is attributable to the general factor.

2.4 | Missing Data

There were a few missing data points at the item level: 0.3%–1.7% of patients did not answer a specific item. Item 9 had the highest number of missing values at T1 (18), and Item 10, at T2 (12). For the total scores, which were calculated according to the official scoring rules that allow for three missing values at the item level, 0.05% missing values occurred. Such low percentages of missing values did not pose a problem for our analyses, nor could we discover a clear missing data pattern. There are also some data missing at random, which are related to the demographic information, shown in Table 1. The numbers are therefore exact, whereas the proportion is based on valid percentages.

2.5 | Statistical Analysis

The main focus of this study is on longitudinal measurement invariance testing using a factor analytic framework to investigate whether the measurement model changes over time. In addition to this analysis, we used IRT modeling to estimate IRT parameters at T1 and Mokken scale analysis (MSA; Mokken 1971; van der Ark 2007) to check whether the crucial

assumption of monotonicity (increasing item response function) holds for all items. Note that the assumption of monotonicity is relevant for both factor analysis (FA) and IRT. A violation of this assumption may lead to a misfit in both FA and IRT models.

At this point in time, assessing longitudinal measurement invariance within an IRT framework is still not very common because of a range of estimation challenges related to the high latent dimensionality associated with longitudinal models. Therefore, we chose to assess longitudinal measurement invariance using an FA framework (and the associated robust WLSMV estimators, based on DWLS in combination with polychoric correlations, rather than maximum likelihood estimators as are commonly used in IRT). We followed the approach outlined by Luo et al. (2022), which entails that a configural model is estimated first, followed by a stepwise procedure to evaluate a series of stricter models and, finally, in case evidence of a breach of invariance is found at the previous step, a set of models is estimated whereby restrictions are lifted iteratively (per item). This procedure enables us to identify which part of the model may violate longitudinal measurement invariance, if any. In a configural model, no restrictions are imposed on the thresholds and loadings: they can freely vary across time points. All item intercepts were constrained to be 0 for model identification purposes.

We will now describe the steps that were taken to subsequently identify which parts of the model are not invariant over time, if any. We start with Step 1, whereby a configural model is estimated in order to determine whether this model has adequate fit; and if so, we proceed to Step 2, whereby an invariant model is estimated with strict restrictions on the threshold and loading—both are held constant over the two time points. If this model fits significantly worse than the configural model, we proceed to Step 3, whereby the restrictions on the threshold and factor loading parameters are removed for one item at a time, and model fit is compared with the invariant model estimated in Step 2. Then, finally, a partial invariance model is estimated in Step 4, whereby item parameters that resulted in a significant improvement in model fit when freed in Step 3 remain free, whereas all other item parameters are held constant over time, resulting in a partial measurement invariance model; which is compared to the configural model in terms of fit. Absolute model fit was evaluated with the widely used comparative fit index (CFI), root mean square error of approximation (RMSEA) and the unbiased standardized root mean square residual (USRMR). We used the rules of thumb often cited in the literature as a guideline for interpretation (i.e., CFA >0.95, RMSEA <0.06 and USRMR <0.08).[3] Relative model fit was evaluated using χ^2 difference tests when comparing models according to the stepwise procedure described above. The significance level α was corrected for multiple testing (due to the iterative approach of identifying which part of the model shows a breach of measurement invariance).

We used the graded response model (GRM; Samejima 1996) to obtain IRT item parameters and test information functions (TIFs). We refer the interested reader to Paap et al. (2020) for a detailed description of the way a GRM is conceptualized and estimated.

MSA (Mokken 1971; Sijtsma and Molenaar 2002) can be viewed as a nonparametric IRT approach, which poses fewer restrictions on the shape of the data distributions and the shape of the relationship between the latent trait and item category endorsement as compared to parametric approaches such as FA and IRT. For this reason, it is an excellent choice for evaluating a crucial assumption of FA and IRT models, namely that the item response functions (which express the relationship between item scores and latent trait scores) are monotonically nondecreasing. This means that as the position of the latent trait increases, so does the score on a specific item (or at the very least, it stays the same; but it does not decrease). In MSA, rest score groups are used as proxies for the latent trait values. In order to evaluate whether monotonicity holds for all 12 LPFS-BF 2.0 items in our data, plots were generated in the MSA framework and visually inspected for each time point.

2.6 | Statistical Software

All analyses were performed in the open software program R version 4.4.2 (R Core Team 2024). For the longitudinal measurement invariance tests, the R packages *semTools* version 0.5–6 (Jorgensen et al. 2021) and *lavaan* version 0.6–19 (Rosseel 2012) were used. The R code provided by Luo et al. (2022) on their github repository was used as a starting point. The IRT analyses were conducted using the *mirt* R package version 1.37.1 (Chalmers 2012). The monotonicity checks were performed with the R package *mokken* version 3.1.0 (van der Ark 2007, 2012). The R package *psych* version 2.4.6.26 (Revelle 2024) was used to obtain descriptive statistics.

3 | Results

3.1 | Distribution of the LPFS-BF 2.0 Sum Scores Across the Two Time Points

As shown in Table 2, the mean scores varied across items, but did not vary much across time points. Items 2 and 11 have the highest item means at both time points, whereas 9 and 12 have the lowest. Figure 1 shows the sum score distributions at both time points. The distributions look similar across the two time points and appear to roughly follow a normal distribution.

3.2 | Monotonicity Assumption Checks

Visual inspection of the MSA-based item response functions did not show any violations of monotonicity at either time point.

3.3 | IRT Parameters and Test Information

When looking at the discrimination parameters for T1 (Table 3), a few things stand out. Firstly, they are rather small: they range between 0.58 (Item 2) and 1.53 (Item 9). When looking at the threshold parameters (b), we can observe that they cover a sufficiently wide range of the latent trait. However, some estimated thresholds had rather extreme values ($>|2.6|$), and for Item 2 at T1, the b1 threshold could not be properly estimated. In sum,

TABLE 2 | Distribution of LPFS-BF 2.0 responses across the two time points.

	Time Point 1			Time Point 2		
	Mean	SD	missing	Mean	SD	missing
1. I often do not know who I really am.	2.76	0.99	7	2.71	1.01	6
2. I often think very negatively about myself.	3.54	0.67	5	3.37	0.78	3
3. My emotions change without me having a grip on them.	3.09	0.91	6	2.90	0.93	5
4. I have no sense of where I want to go in my life.	2.74	1.03	7	2.64	1.04	4
5. I often do not understand my own thoughts and feelings.	2.84	0.91	12	2.71	0.95	7
6. I often make unrealistic demands on myself.	2.90	0.99	12	2.90	1.01	10
7. I often have difficulty understanding the thoughts and feelings of others.	2.14	0.99	8	2.14	0.93	10
8. I often find it hard to stand it when others have a different opinion.	2.24	0.98	6	2.21	0.97	3
9. I often do not fully understand why my behaviour has a certain effect on others.	1.97	0.96	18	1.90	0.91	5
10. My relationships and friendships never last long.	2.19	1.02	11	2.08	0.99	12
11. I often feel very vulnerable when relations become more personal.	3.18	0.97	10	3.11	0.96	6
12. I often do not succeed in cooperating with others in a mutually satisfactory way.	1.71	0.90	12	1.69	0.91	8
Sum score	31.31	6.06	6	30.35	6.43	5
Mean score	2.61	0.50	6	2.53	0.54	5

Note: LPFS-BF 2.0, Level of Personality Functioning Scale—Brief Form 2.0; missing, number of missing values; SD, standard deviation. $N = 1030$. The scores displayed in this table were calculated using the Dutch/internationally used scoring for the answer categories, ranging from 1 to 4 (rather than 0 to 3 as is typical for the Norwegian scoring). Item text reprinted with permission.

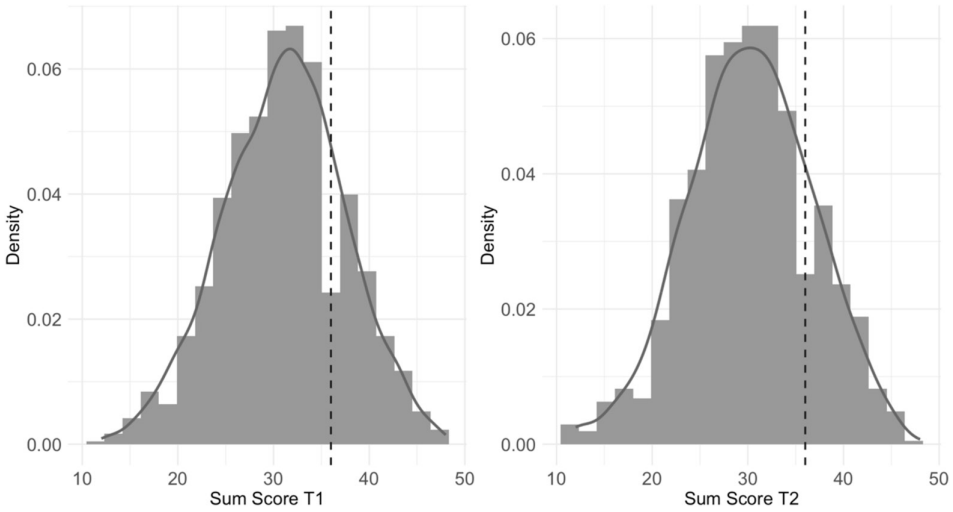


FIGURE 1 | Sum score distribution at Time Point 1 (left panel) and Time Point 2 (right panel). Note. The vertical dashed line indicates the preliminary cutoff suggested by Weekers et al. (2023).

the items do not show high discriminatory power but do cover a sufficiently wide latent trait range. When we look at the TIF for T1 (Figure 2), we notice that there is a clear peak around the mean θ value of 0, with the Information value dropping quite sharply outside of the θ range (−1,1). An information value of 5 corresponds to a local reliability of 0.80 for the type of model we estimated. The value of 5 is just about met for θ values around the mean, but falls below 0.80 for most θ values. The TIF for peak 2 (see Figure S3) looks very similar to that of T1, with the difference that an Information value of 5 is reached for somewhat more respondents.

3.4 | Longitudinal Measurement Invariance Results

We first fitted models that were increasingly strict (in terms of restrictions put on the item parameters) and stopped this

stepwise procedure when we found signs of measurement noninvariance to subsequently find the cause of this noninvariance. We found an acceptable fit for the configural model (the most lenient model we fitted; see Table 4), which allowed us to proceed with comparisons of this model to stricter models. When comparing it with a stricter model whereby thresholds and loadings were kept constant over time, we found a significant deterioration in fit compared to the configural model (Table 5). This implies that we did not find support for strict (scalar) measurement invariance, and we proceeded to investigate which items caused the noninvariance. Because we already found evidence of a breach of invariance, we did not estimate even stricter models.

TABLE 3 | Item parameters based on the unidimensional graded response model at T1.

	Item parameters T1			
Item no.	**a**	**b1**	**b2**	**b3**
1	1.00	−2.30	−0.53	1.18
2	0.58	−8.89[a]	−4.30	−0.98
3	1.06	−3.14	−1.16	0.43
4	0.73	−2.66	−0.60	1.37
5	1.35	−2.32	−0.60	0.98
6	0.67	−3.46	−1.10	1.09
7	1.37	−0.79	0.67	1.88
8	1.28	−1.07	0.52	1.86
9	1.53	−0.45	0.86	2.05
10	1.06	−0.92	0.65	2.12
11	0.86	−3.12	−1.57	0.07
12	1.34	0.09	1.46	2.55

[a]The standard error for this threshold was very large, with 95% CI [−12.41, −5.38]. Note that given a standard normal distribution, threshold values larger than |2.6| can be considered as rather extreme values.

By freeing up constraints for the item parameters across time points, one item at a time, we were able to identify which items were causing the invariance breach. The final model was then a model whereby the constraints across time were lifted for these items, but were kept in place for the items for which measurement invariance held. This model, which we refer to as the 'final partial invariance model', did not significantly differ from the configural model ($p > 0.004$). The estimated absolute fit for this model was acceptable (Table 4). In summary, we can conclude that we found support for partial measurement invariance.

As shown in Table 6, we did not find support for the longitudinal invariance of Items 2, 3 and 7. Item 2 had a somewhat higher factor loading at the second time point, and its threshold parameters were also larger at this time point. For Items 3 and 7, the differences in factor loadings were minimal, but their threshold parameters were notably higher. Specifically, for Item 3, this applied to

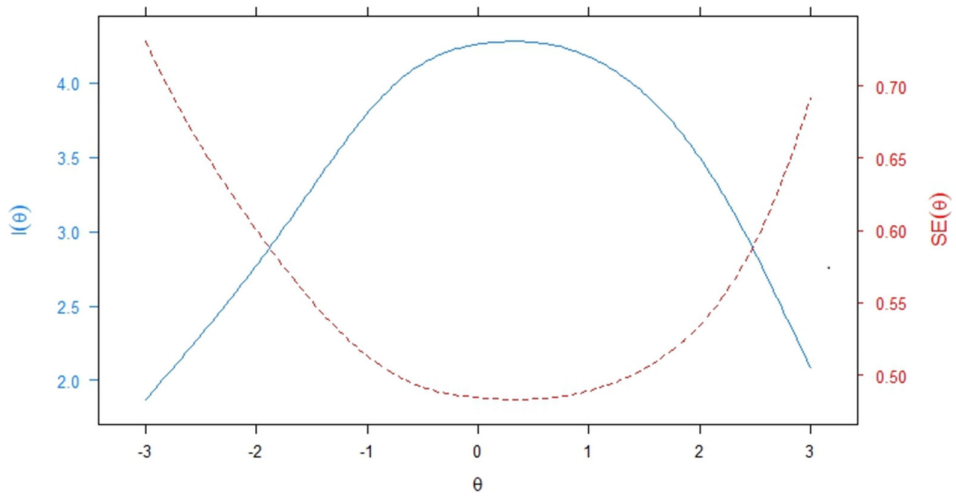


FIGURE 2 | Test information function for T1, with θ on the x-axis, and Fisher information (blue solid line) and the standard error of θ (red dotted line) on the y-axis.

TABLE 4 | Model fit statistics for the estimated longitudinal factor models.

Model	χ^2	χ^2 Scaled	df	CFI	RMSEA	USRMR
Configural model	1678.355	1734.059	239	0.95	0.077	0.075
Strict (scalar) invariance model: Thresholds, loadings, and intercepts	1782.690	1851.003	273	0.95	0.073	0.076
Final partial invariance model	1709.926	1768.560	261	0.95	0.073	0.075

Abbreviations: CFI, comparative fit index; df, degrees of freedom; RMSEA, root mean square error of approximation; USRMR, unbiased standardized root mean square residual.

Threshold 3, whereas for Item 7, it concerned Thresholds 1 and 2. Additional significance testing (data not shown) indicated that the breach of measurement invariance could be attributed to the thresholds.

In order to establish whether breach of measurement invariance had a clinically relevant impact on estimated factor scores, we calculated the absolute mean difference in estimated factor scores for the invariant model, whereby the item parameters were constrained to be equal over time compared to those for the final partial invariance model. The differences we found were very small (mean absolute difference of 0.009 at T1 and 0.025 at T2). This implies that although we did not find support for strict (scalar) measurement invariance, the impact of the noninvariance is in fact negligible. The distributions of the estimated factor scores and the difference scores are presented in the Supporting Information.

TABLE 5 | χ^2 difference tests for fitted models.

Model comparison		χ^2 Diff	df	*p*	*N*
Configural	Versus strict invariance model: Thresholds, loadings and intercepts	126.968	34	< 0.001	1030
Configural	Versus final partial invariance model	37.396	22	0.021	1030

Note: Missing data handling takes into account the full data. χ^2 diff, scaled chi-squared difference test with method = 'satorra.2000'; df, degrees of freedom for the test. A Bonferroni correction was applied in the partial invariance model building, $\alpha_{adjusted} = 0.004$.

4 | Discussion

In this study, we examined the longitudinal measurement invariance of the Norwegian version of the LPFS-BF 2.0 in a large clinical sample. In addition, we estimated and evaluated IRT parameters and test information functions in order to further investigate the usefulness of this instrument in a clinical setting. Our results indicated that the factor structure did not change over time, and measurement invariance across time points held for the majority of the items but did not hold for the entire instrument. Although the latter could be cause for concern, further examination steps showed that the impact of this breach of measurement invariance did not have a substantial impact on the estimated latent trait scores. Strikingly, IRT analyses revealed that the discrimination parameters were much lower than might be expected for a clinical sample, and the test information functions showed that adequate reliability (> 0.80) was only obtained for a narrow range of the latent trait.

The discrimination parameters we estimated were low both in relative and absolute terms. They were considerably lower than the discrimination parameters reported by Weekers et al. (2023) for the Danish general population. In absolute terms, a value of 1 would be considered normal in an educational measurement setting with dichotomous items, but typically much larger values (around 3–4) are found in clinical settings with polytomous items (e.g., Paap et al. 2019; Reise and Waller 2009). A potential explanation for our low discrimination parameters could lie in the nature of the sample. Our

TABLE 6 | Loading and threshold parameter estimates for the final partial invariance model.

	Time Point 1				Time Point 2			
Item	**Loading**	**Thr1**	**Thr2**	**Thr3**	**Loading**	**Thr1**	**Thr2**	**Thr3**
1	0.65	−1.36	−0.34	0.67	0.65	−1.36	−0.34	0.67
2	**0.37**	**−2.63**	**−1.47**	**−0.36**	**0.51**	**−2.42**	**−1.29**	**−0.18**
3	**0.64**	**−1.90**	**−0.74**	**0.27**	**0.70**	**−1.92**	**−0.65**	**0.50**
4	0.48	−1.14	−0.21	0.58	0.48	−1.14	−0.21	0.58
5	0.85	−1.89	−0.45	0.82	0.85	−1.89	−0.45	0.82
6	0.38	−1.34	−0.48	0.39	0.38	−1.34	−0.48	0.39
7	**0.80**	**−0.63**	**0.56**	**1.51**	**0.73**	**−0.86**	**0.44**	**1.52**
8	0.67	−0.85	0.39	1.37	0.67	−0.85	0.39	1.37
9	0.77	−0.39	0.77	1.76	0.77	−0.39	0.77	1.76
10	0.62	−0.57	0.42	1.34	0.62	−0.57	0.42	1.34
11	0.50	−1.55	−0.79	0.04	0.50	−1.55	−0.79	0.04
12	0.76	0.06	1.14	1.96	0.76	0.06	1.14	1.96

Note: Thr, threshold. Intercepts of items are constrained to 0. Parameter estimates are unstandardized. A Bonferroni correction was applied in the partial invariance model building, $\alpha_{adjusted} = 0.004$. The standard errors for the loadings ranged from 0.04 to 0.05 at both time points and from 0.04 to 0.15 for the thresholds across both time points. The highest standard error (0.15) was found for the first threshold of Item 2 at Time Point 1. Items for which longitudinal measurement invariance was not supported are printed in bold.

sample consists of patients who were referred for treatment that addresses personality problems. Focusing on such a specific sample may result in a restriction of range effect (less variation in scores can lead to lower correlations and discrimination parameters). Although the number of PD criteria endorsed (interquartile range: 8–15; median: 11) suggests considerable heterogeneity in PD severity according to the traditional DSM system, the variation in severity according to the AMPD might be less pronounced.

Another effect that could be at play regarding the low discrimination parameters and narrow range of trait values that were measured reliably is that the instrument might simply be too short to adequately capture the richness of the LPFS, which would be especially relevant in a clinical sample. For instance, Item 2 states, 'I often think very negatively about myself'. Although negative self-thoughts are common among both psychiatric patients and the general population, by solely focusing on the frequency of such thoughts, the item may fail to capture the complex and nuanced nature of self-esteem as intended within the LPFS framework. Rather than counting the occurrence of negative self-perception, the LPFS emphasizes that self-esteem is influenced by social interactions, such as exaggerated concern about external evaluation (Level 2), fragile self-esteem easily influenced by events (Level 3) and distorted self-image easily threatened by interactions with others (Level 4). In this respect, Morey's more extensive LPFS-Self-Report (LPFS-SR; Hopwood et al. 2018), consisting of 80 items, may provide a more comprehensive assessment. The LPFS-SR includes at least one item for each severity level across all 12 subdomains of the LPFS, thereby allowing for a more detailed and differentiated evaluation of personality functioning. Perhaps, by placing greater emphasis on the qualitative differences that characterize each level of personality functioning, it might be possible to obtain more reliable scores across the full spectrum of severity—not only in the middle range of the latent scale. Future research directly comparing the psychometric properties, such as reliability and predictive validity, and the utility of the LPFS-BF 2.0 to those of the LPFS-SR in clinical samples would be a welcome addition to the literature.

4.1 | Conclusion and Future Directions

In a recent study by our research group, we found moderate to good support for the use of total scores of the LPFS-BF 2.0 in a large clinical sample (Paap et al. 2024). The current results further strengthen this finding: Although some measurement noninvariance was found, the impact on estimated latent trait scores was very small. In a recent study based on a Danish general population sample, support was found for a unidimensional LPFS-BF 2.0 scale, as well (Weekers et al. 2023). Furthermore, Weekers et al. (2019) reported that the LPFS-BF 2.0 showed a high sensitivity to change after 3 months of treatment in a Dutch patient population. Although these two types of findings, i.e., support for longitudinal measurement invariance and reduction of total LPFS-BF 2.0 scores during therapy, may indicate that the LPFS-BF 2.0 is able to detect change in a valid way, the results of our IRT analyses indicate that the instrument may not provide as much information in a population of adults seeking treatment for PDs as perhaps expected. Our estimated discrimination parameters were considerably lower than those reported for the Danish general population, and much lower than what one would expect for an instrument with polytomous items designed to measure a latent variable related to psychological functioning. What this implies is that the items may provide better discrimination between high and low scorers in a general population than in a clinical population referred for treatment of personality disorders. Therefore, we argue that more research in clinical settings is needed before we can draw firmer conclusions with respect to the suitability of the LPFS-BF 2.0 to measure change in personality problems over time. A second important next step would be to correlate change scores on the LPFS-BF 2.0 with change measured with a clinician-related external criterion variable. This could be as simple as having a clinician answer a single question asking them whether the patient has shown clinically relevant improvement at all time points when the LPFS-BF 2.0 was administered. Ideally, however, change as measured using the LPFS-BF 2.0 would be correlated with change as measured by a diagnostic interview such as the SCID-5-AMPD-I (Bender et al. 2018) or the STiP 5.1 (Hutsebaut et al. 2017). This would allow us to study whether change in reported scores on the LPFS-BF 2.0 is in fact associated with the external criterion 'clinically relevant change' as assessed by a clinician.

Acknowledgements

We wish to thank the patients and staff from the Norwegian Network for Personality Disorders for their contribution to this study. The collaboration includes the following 18 units: Unit for Group Therapy, Øvre Romerike District Psychiatric Center, Akershus University Hospital, Jessheim; Group Therapy Unit, Nedre Romerike District Psychiatric Center, Akershus University Hospital, Lillestrøm; Group Therapy Unit, Follo District Psychiatric Center, Akershus University Hospital, Ski; Group Therapy Unit, Groruddalen District Psychiatric Center, Akershus University Hospital, Oslo; Group Therapy Unit, District Psychiatric Center, Ålesund Hospital, Ålesund; Clinic for Personality disorders, Outpatient Clinic for Specialized Treatment of Personality Disorders, Section for Personality psychiatry and specialized treatments, Oslo University Hospital, Oslo; Group Therapy Unit, Lovisenberg District Psychiatric Center, Lovisenberg Hospital, Oslo; Group Therapy Team, Vinderen Psychiatric Center, Diakonhjemmet Hospital, Oslo; Unit of Personality Psychiatry, Vestfold District Psychiatric Center, Sandefjord; Unit for Intensive Group Therapy, Aust-Agder District Psychiatric Center, Sørlandet Hospital, Arendal; Group Therapy Unit, Kongsvinger District Psychiatric Center, Akershus University Hospital, Kongsvinger; Unit for Group Therapy, District Psychiatric Center, Strømme, Sørlandet Hospital, Kristiansand; Group Therapy Unit, Stavanger Community Mental Healtcare Centre, Stavanger University Hospital, Stavanger; Section for group treatment, Kronstad District Psychiatric Center, Haukeland University Hospital, Bergen; MBT Team, Department of Substance Abuse Medicine, Haukeland University Hospital, Bergen; MBT-Team, Outpatient Clinic, Rogaland A-senter, Stavanger; VPP/Team Sørslettveien, Clinic of Mental Health and Addiction, Northern Norway University Hospital, Tromsø; and Voss District Psychiatric Center—NKS Bjørkeli.

Ethics Statement

Anonymized data were collected from each treatment unit and subsequently transferred to a common research database. The Data Protection Officer (DPO) at each contributing unit approved the data collection procedures, whereas the DPO at Oslo University Hospital, Oslo, Norway, approved the security procedure for the research database.

Consent

All participating patients from all treatment units provided written consent for the anonymous use of their clinical data for research purposes.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Because of restrictions imposed by the Regional Medical Ethics Committee regarding patient confidentiality, data are only available upon reasonable request. Requests for data may be sent to the Privacy and Data Protection Officer at Oslo University Hospital in Oslo, Norway.

Endnotes

[1] Note that there is some overlap in the dataset we used and the data used for another paper by our research team (Paap et al. 2024), where we explored the dimensionality of the instrument at the first time point. Data collection continues to this day, so that the sample size at the various time points continues to increase.

[2] Note that the Norwegian scoring was used in the paper by Paap et al. (2024). That means that, in order to obtain estimates that can be directly compared to those reported here, one would need to add a value of 1 to the LPFS-BF2.0 means reported in Table 3 in the paper by Paap et al. (2024).

[3] It should be acknowledged that these rules of thumb were originally developed for continuous data, and they might be somewhat too lenient in a setting where categorical data are used (Xia and Yang 2019). In line with suggestions made by van Laar and Braeken (2021), we place a greater emphasis on relative fit indices, and the theoretical model is weighed carefully when interpreting the absolute fit measures.

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